

Revisiting Metaheuristic Comparisons for Continuous Wind Farm Layout Optimization: Baseline Configuration, Component Analysis and a Two-Phase PSO-VNS Design

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Abstract—The wind farm layout optimization problem (WFLOP) places a prescribed number of turbines to minimize wake loss under boundary and spacing constraints. This paper shows that conclusions of metaheuristic comparisons for the continuous WFLOP depend strongly on baseline configuration and comparison control. Eight methods are compared on 68 benchmark cases under a controlled protocol: equal budgets including gradient calls, 30 seed-paired runs, feasibility-aware ranking and Holm-corrected tests. First, the particle swarm optimization (PSO) setting $w = 0.7$, $c_1 = c_2 = 2$, common in WFLOP comparisons, violates the particle convergence condition; with it, PSO ranks fifth and is feasible in 86.0% of runs, and salp swarm hybrids appear stronger, whereas with the constriction setting PSO ranks second and outperforms all salp swarm methods. Second, a component analysis with a random-sampling control shows that a variable neighborhood search (VNS) phase helps every swarm, but only a convergent PSO phase gives VNS a better start than random sampling; salp swarm phases and the Laplace step add nothing. Third, the two-phase design, PSO followed by basic VNS (PSO-VNS), obtains the best average rank (1.82), is feasible in every run and exceeds the energy yield of the installed Horns Rev 1 block; however, it is not significantly better than PSO over all cases ($p = 0.30$), VNS alone ranks first from feasible starts, PSO ranks first under a Gaussian wake model, and a gap to the published IEA Wind Task 37 optima remains.

Index Terms—Wind farm layout optimization, particle swarm optimization, variable neighborhood search, benchmarking, metaheuristics, wake effect.

NOMENCLATURE

B	Budget (number of evaluations).
c_1, c_2	Acceleration coefficients of PSO.

C_T	Thrust coefficient.
D, R	Rotor diameter and rotor radius ($D = 2R$).
d_{ij}	Distance of turbine i behind turbine j .
Δ_{ij}, Δ_i	Deficit at i due to j ; total deficit at i .
$E(P_i)$	Expected power of turbine i (benchmark units).
E_{ideal}	Wake-free benchmark objective of one turbine.
F_p	Penalized wake loss (minimized).
g_i^b, g_{ij}^s	Boundary and spacing violations.
$\mathbf{G}, \mathbf{P}^i$	Global best; personal best of particle i .
K	Jensen wake-spreading constant.
$k, k_{\max}$	Index and number of VNS neighborhoods.
$\ell_{ij}, \ell_{\min}$	Distance of i and j ; minimum spacing.
N, N_p	Number of turbines; population size.
N_θ, N_s	Numbers of wind-direction and wind-speed bins.
p_l	Probability of the l th wind-direction bin.
r	Farm radius.
s	Wind speed.
t, T_1	Iteration; number of Phase-1 iterations.
$\mathbf{V}^i$	Velocity of particle i .
w	Inertia weight of PSO.
$\mathbf{X}$	Layout vector $(x_1, y_1, \dots, x_N, y_N)$.
α, β_{ij}	Half-cone angle; wake angle of i behind j .
ζ, ψ	Weibull shape and scale parameters.
θ	Wind direction (blowing towards, CCW from east).
$\kappa, \varphi_{1,2}$	Constriction factor and coefficients.
μ	Penalty factor.
ω	Fraction of the budget given to Phase 1.

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I. INTRODUCTION

The positions of the turbines in a wind farm are fixed at the planning stage and determine, for the lifetime of the farm, how much of the available wind energy it delivers to the grid. An upstream turbine leaves behind a wake of slower and more turbulent air that reduces the power of the turbines downstream, so the layout sets the wake loss and thereby the energy yield on which capacity planning and the design of the collector and grid connection rely [1], [2]. The wind farm layout optimization problem (WFLOP) places a prescribed number of turbines in a site to minimize the wake loss subject to boundary and minimum-spacing constraints. The objective is nonlinear, the wakes couple the positions of all turbines, and the feasible region is bounded by many nonconvex spacing

constraints, which has made WFLOP a popular test bed for metaheuristics.

Early studies placed turbines on a grid with genetic algorithms (GAs) [3], [4]. The continuous, wind-distribution-based model of Kusiak and Song [5], which we also use, has been solved with ant colony optimization and particle filtering [6], [7] and biogeography-based optimization [8]; GA, particle swarm optimization (PSO), differential evolution (DE) and simulated annealing were compared in [9], and PSO has been used to reduce wake losses [10]. Later work used informed GAs [11], [12], deterministic refinement [13], variable neighborhood search (VNS) for large offshore farms [14], pseudo-gradients [15], Gaussian wake models [16] and surrogates [17]–[19]. The IEA Wind Task 37 case study [20] showed that, even with a common wake model, different optimizers reach clearly different layouts.

Many of these studies propose a new or modified metaheuristic and show that it outperforms a set of reference methods. The benchmarking literature has long warned that such comparisons are hard to interpret [21], [22], and four issues matter for the WFLOP. First, the baselines may be configured outside their working range: the particles of PSO converge only if the inertia weight w and the acceleration coefficients c_1 , c_2 lie in a known region [23], [24], and the setting $w = 0.7$, $c_1 = c_2 = 2$, common in PSO baselines and used in the earlier version of this study, lies outside it. Second, budgets counted in iterations, or without the gradient calls of gradient-based solvers, are not equal. Third, the metaheuristics handle the constraints by penalties and may return infeasible layouts, which must be ranked consistently. Fourth, conclusions drawn from mean values without paired tests and multiplicity correction [25], [26] may not hold. For two-phase hybrids, a further question is whether the first phase contributes more than a random start would.

This paper revisits the comparison of metaheuristics for the continuous WFLOP under a protocol that controls these factors. Our thesis is that the conclusions of such comparisons depend strongly on how the baselines are configured and how the comparison is controlled, and that under a controlled protocol a simple two-phase design, a convergent PSO followed by the basic VNS [27], [28] (PSO-VNS), is the most consistent performer, whereas the phases of the salp swarm algorithm (SSA) [29] and the Laplace step of the Laplacian SSA (LX-SSA) [30], proposed earlier by two of the authors, add nothing beyond random sampling. With the old PSO setting, PSO ranks fifth of eight methods and is feasible in 86.0% of the runs, and salp-swarm hybrids appear stronger; with the constriction coefficients of Clerc and Kennedy [23], PSO ranks second and ahead of every salp-swarm method, including an SSA-VNS hybrid that we had developed first. We report this as a property of the setting, not as a finding about any particular study. Our contributions are as follows:

- 1) A controlled evaluation protocol for the continuous WFLOP: equal budgets that count every objective and gradient call, seed-paired runs, feasibility-aware ranking, case-mean and run-level tests with Holm correction, an ablation with a random-sampling control, re-evaluation under alternative modeling assumptions, and public code

and data. We apply it to eight methods on 68 benchmark cases, a 16-turbine block of the Horns Rev 1 offshore farm and IEA Wind Task 37 Case Study 1 (IEA37) (Sections V, VIII and IX).

- 2) Evidence that the PSO baseline setting reverses the ranking of the methods: the setting $w = 0.7$, $c_1 = c_2 = 2$ violates the convergence condition of PSO, and replacing it by the constriction setting moves PSO from behind salp-swarm hybrids to ahead of every salp-swarm method (Sections III-A and VI-A, Table I).
- 3) A component analysis of two-phase swarm-VNS hybrids: the VNS phase helps every swarm, but the swarm phase helps only if the swarm is effective. The SSA and LX-SSA phases are statistically tied with random sampling (RS-VNS) and the Laplace step gives no gain, whereas a PSO phase beats random sampling, VNS alone and the salp-swarm phases (Section VII).
- 4) PSO-VNS as a simple reference method, with an account of where it wins and where it does not. It has the best average rank (1.82), is feasible in every run, keeps the best rank at all three budgets, and its best Horns Rev 1 layout exceeds the energy production of the installed one; but it is not significantly better than PSO over all cases ($p = 0.30$), VNS alone ranks first with feasible starts, PSO ranks first under a Gaussian wake model, and on IEA37 a gap to the published optima remains (Sections VI–IX). From these results we derive practical recommendations for benchmarking WFLOP methods (Section X).

Sections II–IV describe the model, the swarm methods and PSO-VNS, and Section V defines the evaluation protocol. Section VI reports the effect of the baseline setting and the overall comparison, Section VII the component analysis, Section VIII the tests beyond the benchmark and Section IX the sensitivity to the modeling assumptions. Section X gives the recommendations, and Sections XI and XII the limitations and conclusions.

II. PROBLEM MODELING

For a prescribed number N of identical turbines, the $2N$ coordinates (x_i, y_i) , measured from the center of the farm, are optimized to maximize the expected farm power, i.e., to minimize the wake loss, under the Jensen wake model [31], [32] and the power curve of [5]. The wind direction θ has the probability density p_θ , and for each direction the wind speed s follows a Weibull distribution with density $p_s(s; \zeta, \psi) = (\zeta/\psi)(s/\psi)^{\zeta-1} \exp[-(s/\psi)^\zeta]$, shape $\zeta(\theta)$ and scale $\psi(\theta)$. Here θ is the direction *towards* which the wind blows, counter-clockwise from east; the meteorological directions θ_{met} of Horns Rev 1 and IEA37 (wind *from*, clockwise from north) are converted by $\theta = 270^\circ - \theta_{\text{met}}$. Two constraints apply: every pair of turbines satisfies $\ell_{ij} = \|(x_i - x_j, y_i - y_j)\|_2 \geq \ell_{\min} = 4D = 8R$, as in [5], [8] (Section IX discusses $5D$ and $6D$), and the farm is circular, $x_i^2 + y_i^2 \leq r^2$, with $r = 500$, 750 or 1000 m.

A. Wake Effect Model

In the Jensen model, the wake of turbine j is a half cone with half-angle $\alpha = \arctan K$, with the wake-spreading constant K , and its virtual vertex lies R/K upstream of the rotor. With $u_{ij} = (x_i - x_j) \cos \theta + (y_i - y_j) \sin \theta + R/K$ and the lateral offset from the wake axis $v_{ij} = -(x_i - x_j) \sin \theta + (y_i - y_j) \cos \theta$, turbine i lies in the wake if $\beta_{ij} = \cos^{-1}(u_{ij}/\sqrt{u_{ij}^2 + v_{ij}^2}) < \alpha$. Its downstream distance is $d_{ij} = |u_{ij} - R/K|$, and with the thrust coefficient C_T the velocity deficit is

$$\Delta_{ij} = 1 - \frac{s_{\text{down}}}{s_{\text{up}}} = \frac{1 - \sqrt{1 - C_T}}{(1 + K d_{ij}/R)^2}. \quad (1)$$

The deficits are combined by the root-sum-square rule,

$$\Delta_i(\theta) = \left(\sum_{j \neq i, \beta_{ij} < \alpha} \Delta_{ij}^2 \right)^{1/2}. \quad (2)$$

Within a direction bin, the local wind speed is the free-stream speed times $1 - \Delta_i(\theta)$. Since $aS \sim \text{Weibull}(\zeta, a\psi)$ for $S \sim \text{Weibull}(\zeta, \psi)$ and $a > 0$, it keeps the shape parameter and has the scale parameter

$$\psi_i(\theta) = \psi(\theta) [1 - \Delta_i(\theta)], \quad i = 1, \dots, N. \quad (3)$$

Schematics of the farm, the direction convention and the wake cone are in Figs. S12–S14 of the supplementary material.

B. Power Model

The power curve of the Kusiak–Song benchmark [5] is $f(s) = 0$ for $s < s_{\text{cut-in}}$, $f(s) = \lambda_1 s + \lambda_0$ for $s_{\text{cut-in}} \leq s \leq s_{\text{rated}}$ and $f(s) = P_{\text{rated}}$ above, with $s_{\text{cut-in}} = 3.5$ m/s, $s_{\text{rated}} = 14$ m/s, $P_{\text{rated}} = 1500$ kW, $\lambda_1 = 140.86$ and $\lambda_0 = -500$. We use it as published: it is slightly negative just above the cut-in speed, jumps from 1472 to 1500 kW at the rated speed and has no cut-out speed. The expected power of turbine i ,

$$E(P_i) = \int_0^{360} p_\theta(\theta) \int_0^\infty f(s) p_s(s; \zeta(\theta), \psi_i(\theta)) ds d\theta, \quad (4)$$

is evaluated, as in [5], by a Riemann sum over N_θ direction bins with probabilities p_l and N_s speed bins between $s_{\text{cut-in}}$ and s_{rated} (Section S-VI.A of the supplementary material). As in the earlier studies, the direction sum keeps the bin width in degrees as a factor; with 24 bins of 15° and $\sum_l p_l = 1$, the benchmark objective $\sum_i E(P_i)$ is thus 15 times the expected farm power in kW, and $\text{AEP [MWh]} = (\text{objective}/15) \times 8.76$; e.g., the wake-free benchmark objective of one turbine under Wind Data Set I, $E_{\text{ideal}} = 14045.74$, corresponds to 936.38 kW or 8202.7 MWh per year.

C. Constraint Handling

After every position update, each coordinate is clipped to $[-r, r]$ (box repair); the boundary and spacing constraints are handled by a penalty on the violations $g_i^b = x_i^2 + y_i^2 - r^2$ and $g_{ij}^s = \ell_{\min} - \ell_{ij}$, and a layout is feasible if all of them

are ≤ 0 . All penalty-based optimizers minimize the penalized wake loss

$$F_p(\mathbf{X}) = \left(N E_{\text{ideal}} - \sum_{i=1}^N E(P_i) \right) + \sum_{i: g_i^b > 0} (1 + \mu g_i^b)^2 + \sum_{i < j: g_{ij}^s > 0} (1 + \mu g_{ij}^s)^2, \quad \mu = 10^{10}, \quad (5)$$

where $\mathbf{X} = (x_1, y_1, \dots, x_N, y_N)$; since $N E_{\text{ideal}}$ is the wake-free benchmark objective, the first term is the wake loss reported in our tables. Any violation above about 10^{-7} (m or m^2) makes the penalty exceed the largest possible wake loss, so comparing F_p values is a feasibility-first rule: a feasible layout always beats an infeasible one, feasible layouts are compared by wake loss, and infeasible ones by total penalty. The comparison of final runs follows the same order (feasibility-aware ranking, Section V).

III. PRELIMINARIES: PHASE-1 METHODS AND BASIC VNS

This section describes the Phase-1 population methods and the basic VNS. A population method keeps N_p candidate vectors $\mathbf{X}^i \in \mathbb{R}^n$ with components X_m^i and bounds lb_m , ub_m , and it costs N_p objective calls per iteration unless stated otherwise.

A. PSO and Its Convergence Condition

We use the standard global-best PSO [33] with unmodified update rules. Particle i has a position $\mathbf{X}^i$, a velocity $\mathbf{V}^i$ and a personal best $\mathbf{P}^i$, the swarm keeps the global best $\mathbf{G}$, and in the inertia-weight form [34] a particle moves by

$$\mathbf{V}^i \leftarrow w \mathbf{V}^i + c_1 \rho_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2 \rho_2 \circ (\mathbf{G} - \mathbf{X}^i), \quad (6)$$

$$\mathbf{X}^i \leftarrow \mathbf{X}^i + \mathbf{V}^i, \quad (7)$$

where ρ_1, ρ_2 are vectors of independent $U(0, 1)$ numbers, drawn anew for every move and coordinate, and $\circ$ is the componentwise product. Clerc and Kennedy [23] multiply the whole velocity update by the constriction factor $\kappa = 2/|2 - \varphi - \sqrt{\varphi^2 - 4\varphi}|$, $\varphi = \varphi_1 + \varphi_2 > 4$, where φ_1, φ_2 take the place of c_1, c_2 and the inertia weight is 1. Their standard choice $\varphi_1 = \varphi_2 = 2.05$ gives $\kappa = 0.7298$, which is mathematically identical to (6) with

$$w = 0.7298, \quad c_1 = c_2 = \kappa \varphi_1 = 1.49618. \quad (8)$$

Throughout the paper, PSO denotes (6), (7) with these coefficients, and the *old setting* the same algorithm with $w = 0.7$, $c_1 = c_2 = 2$; only the coefficient values differ. The velocities start at zero and are not clamped, the positions are clipped to the bounding square after each move (box repair), and $\mathbf{P}^i$ and $\mathbf{G}$ are updated after each particle evaluation (asynchronous update).

When the personal and global bests stay fixed (stagnation), the mean and the variance of the particle positions converge if $|w| < 1$ and [24]

$$c_1 + c_2 < \frac{24(1 - w^2)}{7 - 5w}. \quad (9)$$

For the constriction setting (8), the right-hand side equals 3.35 and $c_1 + c_2 = 2.99$ satisfies (9). The old setting, which is common in PSO baselines for the WFLOP and was used in the earlier version of this study, violates it: for $w = 0.7$ the bound is 3.50, whereas $c_1 + c_2 = 4$. The variance of the positions then grows even when the bests no longer change, and the swarm keeps sampling far from its best layouts instead of contracting around them. This is a property of the setting, not of PSO; Section VI-A shows that it changes the ranking of the compared methods.

B. Salp Swarm Algorithm and Its Laplacian Variant

The SSA [29] keeps the best solution found so far as the food source $\mathbf{H}$ and splits the sorted population into leaders and followers. A leader $i \leq N_p/2$ updates each coordinate by

$$X_m^i = H_m \pm \xi_1((ub_m - lb_m)\xi_2 + lb_m), \quad (10)$$

with either sign with probability $1/2$, $\xi_2 \sim U(0, 1)$ and $\xi_1 = 2 \exp[-(4t/T)^2]$ decreasing with the iteration t out of T ; a follower $i > N_p/2$ moves to the midpoint $\frac{1}{2}(X_m^i + X_m^{i-1})$ between itself and the preceding salp. LX-SSA [30] gives each follower a second candidate $Y_m^i = X_m^i + \gamma(H_m - X_m^i)$ with the Laplace step [35]

$$\gamma = \phi - \chi \operatorname{sgn}(z - \tfrac{1}{2}) \ln(1 - 2|z - \tfrac{1}{2}|), \quad z \sim U(0, 1), \quad (11)$$

location $\phi = 0$ and scale $\chi = 1$, and keeps the better candidate, so that one iteration costs $2N_p$ calls. Comparing SSA with LX-SSA isolates the Laplace step. Pseudocode is given in Section S-I.B of the supplementary material.

C. Basic Variable Neighborhood Search

The basic VNS [27], [28], in the continuous form of [36], improves one incumbent $\mathbf{X}$. It uses $k_{\max} = 5$ nested ℓ_∞ neighborhoods $\mathcal{N}_k(\mathbf{X}) = \{\mathbf{X}' : \delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \leq \delta_k\}$, $\delta_k = 0.1 k r$, of the whole layout. Shaking draws a point uniformly from $\mathcal{N}_k(\mathbf{X})$ (box-repaired), and a complete best-improvement local search improves it. Since the top-hat Jensen wake makes F_p discontinuous, the local search is a derivative-free compass search: it evaluates all $2n$ moves $\pm h \mathbf{e}_m$, moves to the best improving one, and halves h (from $0.05r$) when none improves, until $h < 10^{-3}r$. A better local minimum replaces the incumbent and resets k to 1; otherwise k is increased cyclically. The stand-alone VNS starts from the best member of the initial population.

IV. TWO-PHASE SWARM-VNS DESIGN AND PSO-VNS

A. Two-Phase Template

All hybrids in this paper follow one template. For a budget of B objective calls and a split $\omega \in (0, 1)$, Phase 1 starts from the seeded initial population, runs a population method for about ωB calls and passes its best layout to Phase 2, the basic VNS, which applies the local search to it and then iterates until B calls are used. The instances differ only in Phase 1: PSO in *PSO-VNS*; SSA or LX-SSA in *SSA-VNS* and *LX-SSA-VNS* (ξ_1 decreasing over the Phase-1 iterations); and pure random sampling of $\operatorname{round}(\omega B)$ layouts, uniform

Algorithm 1 PSO-VNS for a prescribed turbine count N

Require: budget B , split ω , N_p , w , c_1 , c_2 , radius r , $k_{\max}$

- 1: Draw and evaluate N_p particles $\mathbf{X}^i$; $\mathbf{V}^i \leftarrow \mathbf{0}$; $\mathbf{P}^i \leftarrow \mathbf{X}^i$;
 $\mathbf{G} \leftarrow \arg \min_i F_p(\mathbf{P}^i)$ ▷ Phase 1
- 2: **for** $t = 1, \dots, T_1$ and $i = 1, \dots, N_p$ **do**
- 3: Move $\mathbf{X}^i$ by (6), (7); box repair; evaluate $F_p(\mathbf{X}^i)$;
 update $\mathbf{P}^i$, $\mathbf{G}$
- 4: **end for**
- 5: $\mathbf{X} \leftarrow \text{LOCALSEARCH}(\mathbf{G})$; $k \leftarrow 1$ ▷ Phase 2
- 6: **while** fewer than B objective calls have been used **do**
- 7: $\mathbf{X}' \leftarrow \text{LOCALSEARCH}(\text{uniform point of } \mathcal{N}_k(\mathbf{X}))$
- 8: **if** $F_p(\mathbf{X}') < F_p(\mathbf{X})$ **then** $\mathbf{X} \leftarrow \mathbf{X}'$, $k \leftarrow 1$ **else** $k \leftarrow$
 $k \bmod k_{\max} + 1$
- 9: **end while**
- 10: **return** best layout evaluated

in the bounding square and including the initial population, in the control *RS-VNS*. VNS alone is the limiting case in which Phase 1 is the initial population. PSO and SSA run for $T_1 = \operatorname{round}((\omega B - N_p)/N_p)$ Phase-1 iterations, since the initial population costs N_p calls and every iteration another N_p , and LX-SSA for $\operatorname{round}((\omega B - N_p)/(2N_p))$. A swarm phase is useful only if it gives VNS a better start than random sampling with the same number of calls; the component analysis (Section VII) tests this for each Phase-1 choice and, by comparing SSA-VNS with LX-SSA-VNS, isolates the Laplace step.

B. PSO-VNS

PSO-VNS is not a new PSO variant: Phase 1 is the plain PSO of Section III-A, and its final global best $\mathbf{G}$ is the starting point of the basic VNS (Algorithm 1). Both phases minimize F_p (5) over the box-repaired layout vectors $\mathbf{X} \in [-r, r]^{2N}$, so the feasibility-first rules of Section II-C hold in every comparison, best update and acceptance step. We use PSO in Phase 1 because, with coefficients that satisfy (9), the swarm contracts around a good region within its share of the budget, so that VNS starts from a refined layout. Since the coefficients do not depend on the iteration, Phase 1 reproduces exactly, for a given seed, the first $N_p(T_1 + 1)$ calls of a PSO run; PSO-VNS and PSO differ only in how they spend the rest of the budget, and comparing them isolates the use of its second part.

PSO-VNS uses $N_p = 30$, the coefficients (8) and the VNS settings above, and adds one parameter, the split ω . With the default $\omega = 0.5$ and $B = 6,030$, PSO runs for $T_1 = 100$ iterations ($30 + 100 \times 30 = 3,030$ calls) and VNS receives the remaining 3,000. None of these values was tuned; Section VII-A studies ω . Both phases are dominated by the objective evaluations, so PSO-VNS has the same $\mathcal{O}(B N^2 N_\theta N_s)$ cost as the other methods. Phase 1 draws the initial population first, so runs with the same seed start from the same population as all other methods and are paired.

V. EVALUATION PROTOCOL

The protocol below applies to every method and every study.

A. Cases and Methods

The 68 benchmark cases combine the two wind data sets of the Kusiak–Song benchmark [5], [8] (Table S1 of the supplementary material; Data Set I strongly directional, Data Set II spread over more directions; $N_\theta = 24$, $N_s = 21$) with circular farms of radius $r = 500, 750$ and 1000 m and $N = 2, \dots, 10, 2, \dots, 12$ and $2, \dots, 15$ turbines ($R = 38.5$ m, $\ell_{\min} = 4D = 308$ m, $C_T = 0.8$, $K = 0.075$).

All methods minimize F_p (5) over $[-r, r]^{2N}$, with $N_p = 30$ where applicable. The *main comparison* includes eight methods: PSO-VNS (Algorithm 1, $\omega = 0.5$); PSO [23], [33] with (8); SSA-VNS (SSA in Phase 1 of Algorithm 1); SSA [29]; LX-SSA [30] ($\phi = 0$, $\chi = 1$); DE/rand/1/bin [37] (scale factor 0.5, crossover rate 0.9); VNS (Phase 2 alone, from the best initial layout); and MS-SLSQP, SciPy’s SLSQP [38] with explicit constraints and forward-difference gradients (1-m step), restarted from the initial layouts, best F_p first, until the budget is used up. The *component analysis* (Section VII) adds LX-SSA-VNS (LX-SSA in Phase 1) and the random-sampling control RS-VNS, which evaluates $\text{round}(\omega B) = 3,015$ uniform random layouts (including the initial population) and refines the best one by the same VNS, and it varies ω on twelve cases. The *baseline analysis* (Table I) ranks the pool LX-SSA-VNS, SSA-VNS, LX-SSA, SSA, PSO, DE, VNS and MS-SLSQP twice, with the constriction PSO and with the PSO runs with $w = 0.7$, $c_1 = c_2 = 2$ of the earlier version of this study (same code, cases, budget and seeds; Section S-II.A).

B. Comparison Design and Statistics

Equal budgets and paired seeds. Every run of the main comparison and the component analysis has exactly $B = 6,030$ objective calls, counted by a wrapper around the objective function; the initial population and the finite-difference gradient calls of MS-SLSQP count. All iteration counts follow from B (e.g., 200 for PSO, 100 for LX-SSA at $2N_p$ calls each; 3,030 Phase-1 calls in the hybrids). Each method is run with the seeds 1–30, and every optimizer draws its initial population first, so runs with the same seed start from the *same* initial population for all methods and are paired.

Feasibility-aware ranking. A layout is feasible if the spacing and boundary constraints hold within 10^{-6} m; means are taken over the feasible runs. Within a case, a method is ranked by the mean benchmark objective of its feasible runs only if at least 15 of its 30 runs (5 of 10) are feasible, and otherwise below all such methods, by its number of feasible runs, so that a few favorable feasible runs cannot earn a good rank.

Statistics. All tests are two-sided with $\alpha = 0.05$ and Holm’s correction [39]. 1) Case ranks: the Friedman test with the Iman–Davenport correction and post hoc tests of the average ranks against PSO-VNS [25]. 2) Case means: since mean-rank post hoc tests depend on the other methods in the pool [26], Wilcoxon signed-rank tests of PSO-VNS against each method on the 68 per-case mean wake losses. 3) Runs: within each case, Wilcoxon signed-rank tests on the 30 seed-paired runs (every infeasible run ranked below every feasible run), adjusted over the comparisons of the case and counted as significant wins, ties and significant losses (W/T/L), with

the matched-pairs rank-biserial correlation r_{rb} as effect size. The component analysis applies 3) to planned contrasts.

C. Tests Beyond the Benchmark and Reproducibility

Initialization and budget. Random initial layouts (uniform in the bounding square) are usually infeasible; the feasible initialization draws each initial layout inside the site and pushes the turbines apart with L-BFGS-B on a smooth packing penalty, without objective calls. The nine methods other than RS-VNS are run with it at 6,030 calls; RS-VNS would need one packing solve for each of its 3,015 random layouts, which costs far more than the objective calls. With random initialization, all ten methods are run at 6,030, 30,030 and 120,030 calls. Both studies use the six largest cases (largest N of each farm and data set) and the Horns Rev 1 block, with 30 seeds (10 for Horns Rev 1 at 120,030 calls).

Other sites and models. The Horns Rev 1 block is PyWake’s 16-turbine example [40]: V80 turbines (2 MW, $D = 80$ m, 25-m/s cut-out), the measured 12-sector wind climate, the parallelogram through the corner turbines as boundary, $\ell_{\min} = 4D$, and the Jensen wake with $K = 0.04$. The 16- and 36-turbine scenarios of IEA37 Case Study 1 [20], [41] have circular boundaries of radius 1300 and 2000 m, $\ell_{\min} = 2D = 260$ m, 16 directions at 9.8 m/s and a simplified Bastankhah Gaussian wake. All ten methods minimize the penalty (5) on the AEP loss at 6,030 and 30,030 calls (30 seeds), and their layouts are compared with the submitted ones. For the robustness checks, every feasible final layout of the main comparison is re-evaluated, not re-optimized, with a cubic power curve (with and without cut-out) and a Gaussian wake model (Section IX).

Verification and code. Our objective agrees with the original implementation within a relative difference of 10^{-6} (720 archived final objectives within 8.7×10^{-11}), with PyWake within 0.3% of the Horns Rev 1 AEP, and with the official IEA37 calculator within a relative 10^{-11} . SSA, LX-SSA, PSO and DE (also in the hybrids) use our original code, which reproduces the archived result distributions of the earlier version (22 of 24 Mann–Whitney tests with $p \geq 0.05$); the other methods were implemented for this study. The runs used Linux containers (Intel Xeon, 2.1 GHz, 4 virtual cores; Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1). The code, all per-run results and the scripts behind every table, figure and quoted number are public (see Data Availability).

VI. RESULTS ON THE BENCHMARK

This section reports the 68 benchmark cases (20,400 runs): the effect of the PSO baseline setting, the overall comparison, and the quality, feasibility and convergence of the solutions; the component analysis follows in Section VII. Per-case means and standard deviations are given in Tables S2–S7 of the supplementary material.

A. Effect of the Baseline Setting

The setting $w = 0.7$, $c_1 = c_2 = 2$, common in PSO baselines for the WFLOP and used in the earlier version of this study, violates the convergence condition (9). We ran PSO

TABLE I

EFFECT OF THE PSO SETTING ON THE COMPARISON OF THE PREVIOUS STUDY'S METHODS (68 CASES, 6,030 EVALUATIONS): AVERAGE RANK (1 = BEST) WITH THE OLD ($w = 0.7$, $c_1 = c_2 = 2$) AND THE CONSTRICTION SETTING, AND RUN-LEVEL W/T/L OF THE SALP-SWARM HYBRIDS AGAINST PSO

Method	PSO setting	
	Old	Constriction
LX-SSA-VNS	2.65	3.41
SSA-VNS	1.99	2.77
LX-SSA	5.65	6.06
SSA	4.81	5.34
PSO	5.04	1.53
DE	6.99	7.03
VNS	3.59	4.15
MS-SLSQP	5.28	5.71
PSO position	5	1
LX-SSA-VNS vs. PSO (W/T/L)	35/33/0	0/21/47
SSA-VNS vs. PSO (W/T/L)	41/27/0	0/22/46

Ranks: feasibility-aware rule (fewer than 15 of 30 feasible runs in a case = ranked last); bold: best average rank. W/T/L: cases in which the hybrid is significantly better / not different / worse than PSO (two-sided Wilcoxon signed-rank test, 30 seed-paired runs, Holm-adjusted over the seven comparisons of the hybrid in each case, $\alpha = 0.05$).

with this setting on the same cases with the same budget and seeds. Compared with the constriction setting (8), it returns a feasible layout in only 86.0% of the runs (fewer than 15 feasible runs in 8 cases), is significantly worse in 56 of the 68 cases and never better (Wilcoxon, 30 seed-paired runs), and has a mean wake loss 0.61 percentage points higher over the 60 cases in which both settings have at least 15 feasible runs. Among the eight methods of Table II, it would place PSO fifth (average rank 5.12) instead of second, where PSO ranks ahead of SSA, LX-SSA and SSA-VNS.

Table I shows what this does to the conclusions of a comparison. It ranks the method pool of the earlier study (without PSO-VNS) once with each PSO setting, with the same runs of all other methods. With the old setting, SSA-VNS has the best average rank and PSO ranks behind a salp-swarm hybrid (average ranks: LX-SSA-VNS 2.65, SSA-VNS 1.99, PSO 5.04; W/T/L of the two hybrids against PSO 35/33/0 and 41/27/0). With the constriction setting, PSO (1.53) ranks ahead of every salp-swarm method, including both hybrids (3.41 and 2.77; W/T/L 0/21/47 and 0/22/46). Changing the coefficients of one baseline thus reverses the order of PSO and the salp-swarm hybrids.

This is a property of the setting, not of any particular study: a PSO baseline outside the convergence region understates what PSO achieves on the WFLOP, so a method compared with it can appear stronger than it is. PSO baselines should therefore be checked against (9) or use the coefficients (8). On this benchmark, the salp-swarm methods, including our own LX-SSA, do not outperform a correctly configured PSO, which is why PSO forms the first phase of PSO-VNS; all other results use the constriction setting.

B. Overall Comparison

Table II summarizes the comparison of the eight methods. The Friedman test rejects the hypothesis that they perform

TABLE II

CASE-LEVEL ANALYSIS OVER THE 68 BENCHMARK CASES. AVG. RANK: AVERAGE RANK OF THE MEAN FEASIBLE OBJECTIVE (1 = BEST; METHODS WITH FEWER THAN 15 FEASIBLE RUNS IN A CASE RANKED LAST); "BEST": CASES IN WHICH THE METHOD ALONE RANKS FIRST; "FEAS.": FEASIBLE RUNS (%); p_z : HOLM-ADJUSTED p OF THE AVERAGE-RANK TEST AGAINST PSO-VNS. BECAUSE MEAN-RANK TESTS DEPEND ON THE POOL OF METHODS [26], p_W AND $\overline{\Delta L}$ GIVE THE HOLM-ADJUSTED TWO-SIDED WILCOXON SIGNED-RANK TEST ON THE 68 PER-CASE MEAN WAKE LOSSES AND THE MEAN WAKE-LOSS DIFFERENCE (PERCENTAGE POINTS, PSO-VNS MINUS METHOD; NEGATIVE = PSO-VNS BETTER)

Method	Avg. rank	Best	Feas.	p_z	p_W	$\overline{\Delta L}$
PSO-VNS	1.82	29	100.0	—	—	—
PSO	1.97	25	100.0	0.713	0.296	-0.018
SSA-VNS	3.45	0	99.6	2.0×10^{-4}	5.7×10^{-11}	-0.330
VNS	4.40	3	100.0	2.4×10^{-9}	1.3×10^{-10}	-0.465
SSA	5.38	0	97.6	8.3×10^{-17}	4.5×10^{-11}	-0.611
MS-SLSQP	5.88	1	100.0	1.8×10^{-21}	4.2×10^{-11}	-0.435
LX-SSA	6.06	0	96.9	3.3×10^{-23}	4.5×10^{-11}	-0.765
DE	7.04	0	71.4	1.0×10^{-34}	5.6×10^{-11}	-0.887

Friedman $\chi_F^2 = 319.3$ (7 d.f.), $p = 4.6 \times 10^{-65}$; Iman-Davenport $F_F = 136.5$

TABLE III

PAIRWISE OUTCOME OF PSO-VNS AGAINST EACH METHOD: CASES IN WHICH PSO-VNS IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (TWO-SIDED WILCOXON SIGNED-RANK TEST, 30 SEED-PAIRED RUNS, HOLM-ADJUSTED OVER THE SEVEN COMPARISONS OF EACH CASE, $\alpha = 0.05$; AN INFEASIBLE RUN RANKS BELOW EVERY FEASIBLE RUN); LAST ROW: MEDIAN RANK-BISERIAL CORRELATION (POSITIVE = PSO-VNS BETTER)

DS	r (m)	Cases	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
I	500	9	0/8/1	6/3/0	8/1/0	8/1/0	7/2/0	6/3/0	7/2/0
I	750	11	1/8/2	9/2/0	10/1/0	10/1/0	9/2/0	8/3/0	10/1/0
I	1000	14	3/11/0	9/5/0	12/2/0	12/2/0	12/2/0	12/2/0	12/2/0
II	500	9	3/5/1	7/2/0	7/2/0	8/1/0	7/2/0	7/1/1	8/1/0
II	750	11	2/9/0	8/3/0	10/1/0	10/1/0	9/2/0	8/3/0	9/2/0
II	1000	14	5/9/0	10/4/0	12/2/0	12/2/0	12/2/0	10/4/0	13/1/0
All		68	14/50/4	49/19/0	59/9/0	60/8/0	56/12/0	51/16/1	59/9/0
$\bar{r}_{rb}$			+0.08	+0.72	+0.98	+0.97	+1.00	+0.84	+0.99

equally ($\chi_F^2 = 319.3$, $p = 4.6 \times 10^{-65}$). PSO-VNS has the best average rank (1.82) and alone ranks first in 29 cases, followed by PSO (1.97), SSA-VNS (3.45), VNS (4.40), SSA (5.38), MS-SLSQP (5.88), LX-SSA (6.06) and DE (7.04). In the Holm-adjusted post hoc tests, PSO-VNS ranks significantly better than SSA-VNS, VNS, SSA, MS-SLSQP, LX-SSA and DE ($p_{\text{Holm}} \leq 2.0 \times 10^{-4}$), but not than PSO ($p_{\text{Holm}} = 0.71$).

The run-level tests in Table III locate these differences. Against PSO, which forms its first phase, PSO-VNS is significantly better in 14 cases, worse in 4 and tied in 50; its mean wake loss is 1.322% against 1.341%, and the difference of the per-case means is not significant (two-sided Wilcoxon signed-rank test, $p = 0.30$). The difference concentrates in the larger cases: for $N \geq 10$, PSO-VNS wins 9 of the 20 cases and loses 2, with a mean gain of 0.08 percentage points, whereas for $N < 10$ both reach nearly the same wake loss (mean absolute difference 0.03 percentage points). Against SSA-VNS, the same design with a salp-swarm first phase, PSO-VNS is significantly better in 49 cases and never worse (1.322% vs. 1.653%, $p = 1.9 \times 10^{-11}$). Against SSA, LX-SSA and DE, it is significantly better in 59, 60 and 56 cases, and

against VNS and MS-SLSQP in 51 and 59 cases; it is never significantly worse than SSA-VNS, SSA, LX-SSA, DE and MS-SLSQP and significantly worse than VNS in one case.

These results support PSO-VNS as a simple reference method, within limits. Under this protocol it is the most consistent of the eight methods: feasible in every run, at least as good as a correctly configured PSO and better for $N \geq 10$; with up to 20 times the budget, PSO-VNS keeps the best average rank at all three budgets (Section VIII-A). Its gain over PSO, however, is small and not significant over all cases, and the order of the leading methods depends on the test conditions: with feasible initial layouts, VNS has the best average rank (Section VIII-A); under the Gaussian re-evaluation, PSO ranks first (Section IX); and on IEA37 Case Study 1, the best layouts remain below the best published ones (Section VIII-C).

C. Solution Quality, Feasibility and Convergence

The wake loss grows with N , falls as the farm becomes larger, and is higher for the broader wind rose of Data Set II (wake loss versus N for all methods in Section S-II.D of the supplementary material). For $N \leq 3$, all methods stay within 0.25% of the wake-free objective; for 15 turbines in the 1000-m farm of Data Set I, the mean wake loss is 1.60% for PSO-VNS, 1.62% for PSO, 1.69% for SSA-VNS and 2.80% for SSA. Over the six largest cases, the mean wake loss of PSO minus that of PSO-VNS ranges from -0.18 to $+0.41$ percentage points, with PSO-VNS lower in four of them.

PSO-VNS ends with a feasible layout in all runs; the shares of the other methods are given in Table II (e.g., 71.4% for DE). The feasibility counts of PSO, SSA, LX-SSA and DE are the same for both wind data sets, because as long as every candidate is infeasible, the penalty dominates F_p . In the densest cases (500 m, $N = 10$), the global best of the PSO phase is feasible at the switch in only 63.3% of the runs, but all runs are feasible after the VNS phase.

Fig. 1 shows the second phase directly: PSO-VNS and PSO both keep improving after the switch. Over all feasible runs, the VNS phase removes on average 30.3% (median 22.6%) of the wake loss left at the end of the PSO phase and makes 26 runs that were infeasible at the switch feasible, whereas continuing PSO for the same 3,000 calls removes 29.2% (median 20.6%). The VNS phases of SSA-VNS and RS-VNS remove larger shares (36.7% and 57.1%) but start from worse or more often infeasible layouts (feasible at the switch in 98.7%, 95.7% and 66.7% of the PSO-VNS, SSA-VNS and RS-VNS runs).

VII. COMPONENT ANALYSIS OF TWO-PHASE HYBRIDS

The eight variants of Table IV combine a Phase 1 (PSO, SSA, LX-SSA, random sampling or none) with or without the same VNS phase, at 6,030 calls and with the same seeds (convergence in Fig. 1). The Friedman test rejects equal average ranks ($\chi^2_F = 326.8$, 7 d.f., $p = 1.1 \times 10^{-66}$).

What the VNS phase contributes. Adding VNS helps every swarm: each hybrid ranks ahead of the same swarm run alone for the full budget (PSO-VNS vs. PSO, SSA-VNS vs. SSA

TABLE IV
ABLATION OVER THE 68 BENCHMARK CASES (30 SEED-PAIRED RUNS, 6,030 CALLS). TOP: AVERAGE RANK AMONG THE EIGHT VARIANTS (FRIEDMAN $\chi^2_F = 326.8$, 7 D.F., $p = 1.14 \times 10^{-66}$) AND FEASIBLE RUNS. BOTTOM: PLANNED CONTRASTS, CASES IN WHICH THE FIRST VARIANT IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (WILCOXON, HOLM-ADJUSTED OVER THE EIGHT CONTRASTS OF EACH CASE, $\alpha = 0.05$), AND MEAN DIFFERENCE OF THE WAKE LOSS $\Delta \bar{L}$ (PERCENTAGE POINTS; NEGATIVE = FIRST VARIANT BETTER)

Variant	Phase 1	Phase 2	Avg. rank	Feas. (%)
PSO-VNS	PSO	VNS	1.87	100.0
SSA-VNS	SSA	VNS	3.79	99.6
LX-SSA-VNS	LX-SSA	VNS	4.62	99.9
RS-VNS	random sampling	VNS	4.46	99.4
VNS	–	VNS	5.32	100.0
PSO	PSO	–	2.10	100.0
SSA	SSA	–	6.56	97.6
LX-SSA	LX-SSA	–	7.29	96.9

Contrast	Isolates	W/T/L	$\Delta \bar{L}$
PSO-VNS vs. PSO	VNS phase	8/58/2	−0.018
PSO-VNS vs. VNS	swarm start vs. best initial point	50/17/1	−0.465
PSO-VNS vs. RS-VNS	swarm vs. random sampling	48/20/0	−0.399
PSO-VNS vs. SSA-VNS	PSO vs. SSA as Phase 1	47/21/0	−0.330
PSO-VNS vs. LX-SSA-VNS	PSO vs. LX-SSA as Phase 1	48/20/0	−0.393
SSA-VNS vs. RS-VNS	SSA vs. random sampling	2/66/0	−0.068
SSA-VNS vs. LX-SSA-VNS	Laplace step in the hybrid	3/65/0	−0.063
LX-SSA vs. SSA	Laplace step alone	0/61/7	+0.166

and LX-SSA-VNS vs. LX-SSA in Table IV); e.g., the mean wake loss falls from 1.805% (SSA) to 1.653% (SSA-VNS). PSO-VNS and PSO share their first 3,030 calls, so their contrast measures only the use of the second half: switching to VNS instead of continuing the swarm gives a small gain that grows with N (8/58/2 under the Holm adjustment over the contrasts of Table IV, mean difference -0.018 percentage points; Section VI-B): a convergent PSO keeps improving in the second half on small layouts, whereas on larger layouts the single-turbine moves of the compass search pay off.

What the swarm phase contributes. Two controls isolate Phase 1. VNS alone starts from the best initial layout and spends the whole budget on refinement; RS-VNS spends the 3,015 calls of Phase 1 on independent random layouts and refines the best one. PSO-VNS is significantly better than VNS in 50 cases and worse in 1 (-0.465 percentage points), and better than RS-VNS in 48 cases and never worse (-0.399 percentage points). The PSO phase thus adds value beyond a longer refinement and beyond a random multistart of equal cost. The salp-swarm phases do not: SSA-VNS and RS-VNS are statistically tied in 66 of the 68 cases (2/66/0, -0.068 percentage points), and LX-SSA-VNS ranks even behind RS-VNS on average. As start generators for VNS, SSA and LX-SSA are no better than random sampling, which itself ranks ahead of VNS alone.

Laplace step. The Laplace step of LX-SSA gives no gain: without the VNS phase, LX-SSA is 0/61/7 against SSA, and inside the hybrid, SSA-VNS is 3/65/0 against LX-SSA-VNS (-0.063 percentage points).

Choice of Phase 1. Among the Phase-1 options tested, only PSO is effective. With the same VNS phase, PSO-VNS is 47/21/0 (W/T/L) against SSA-VNS and 48/20/0 against LX-SSA-VNS. At the switch, its global best is feasible more

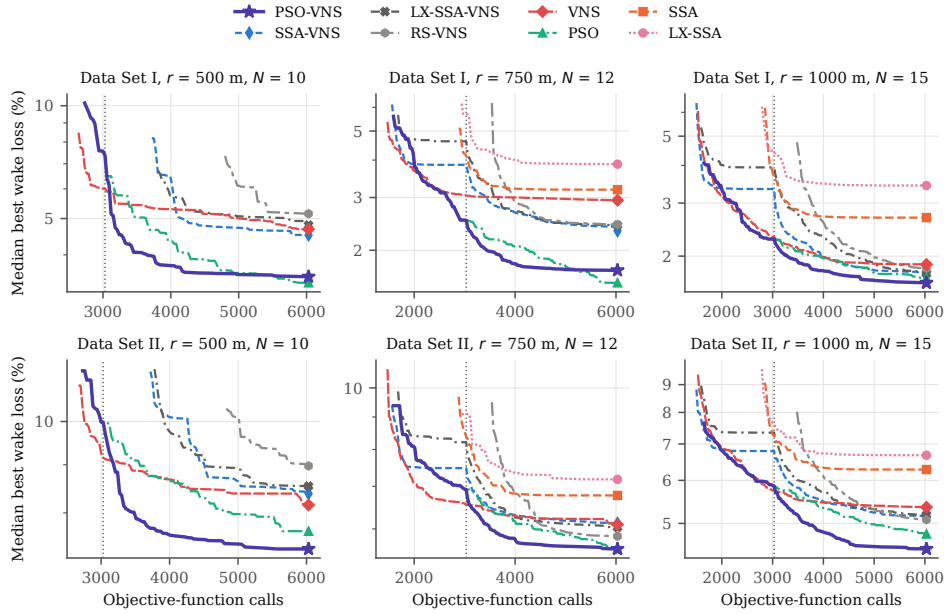


Fig. 1. Convergence of the eight ablation variants (Table IV) for the largest turbine count of each farm: median best feasible wake loss over 30 runs versus objective calls. PSO-VNS and PSO coincide up to 3,030 calls (dotted line), where PSO-VNS switches to VNS. The curves of the eight methods of the main comparison are given in Fig. S3 of the supplementary material.

TABLE V

BUDGET SPLIT OF PSO-VNS (25%, 50% AND 75% OF THE 6,030 CALLS FOR PSO) ON 12 CASES (MIDDLE AND LARGEST N OF EACH FARM AND DATA SET, 30 SEEDS): MEAN WAKE LOSS, AVERAGE RANK AMONG THE THREE SPLITS, AND CASES IN WHICH THE DEFAULT 50% SPLIT IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (WILCOXON, HOLM-ADJUSTED)

PSO share ω	Mean loss (%)	Avg. rank	50% vs. ω (W/T/L)
25%	2.796	2.92	6/6/0
50%	2.643	2.00	—
75%	2.543	1.08	0/7/5

often (98.7% of the runs vs. 95.7% and 94.1%) and, when feasible, has a lower wake loss (1.63% vs. 2.01% and 2.11%), so VNS starts from a better layout. The first phase of a hybrid should therefore be judged against a random-sampling control of equal cost.

A. Budget Split

The split $\omega = 0.5$ was fixed before the experiments and is used for all other results; Table V compares it with $\omega = 0.25$ and 0.75 on the twelve cases with the middle and the largest N of each farm (per-case values in Table S10 of the supplementary material). A longer PSO phase is better: with $\omega = 0.75$, the mean wake loss is lower than with $\omega = 0.5$ in eleven of the twelve cases, significantly in five, and 0.10 percentage points lower on average, and the default is significantly better in no case; $\omega = 0.25$ is significantly worse than the default in six cases and better in no case. In line with the component analysis, most of the benefit comes from the PSO phase, and VNS serves as a final refinement. We do not re-tune ω on the test cases, so the main comparison may slightly understate PSO-VNS; since the split study covers only twelve cases and

TABLE VI

SIX LARGEST BENCHMARK CASES: MEAN WAKE LOSS (%) AND FEASIBLE RUNS (%) AT 6,030 CALLS WITH RANDOM AND FEASIBILITY-PRESERVING INITIALIZATION, AND AVERAGE RANK AT 6,030, 30,030 AND 120,030 CALLS (RANDOM INITIALIZATION; 30 SEEDS); N/R: NOT RUN

Method	Random init.		Feasible init.		Avg. rank		
	Loss	Feas.	Loss	Feas.	6,030	30,030	120,030
PSO-VNS	4.084	100.0	3.938	100.0	1.50	1.17	2.17
SSA-VNS	4.678	95.0	3.955	100.0	4.33	4.17	4.17
LX-SSA-VNS	4.802	98.9	3.952	100.0	5.17	5.67	5.83
RS-VNS	4.830	93.3	n/r	n/r	4.67	6.33	4.67
VNS	4.950	100.0	3.810	100.0	6.50	4.83	3.17
PSO	4.181	100.0	5.549	100.0	1.83	2.50	5.83
SSA	4.828	80.0	4.947	100.0	8.33	8.67	8.83
LX-SSA	5.379	77.8	5.048	100.0	8.67	9.83	10.00
DE	—	1.1	5.549	100.0	10.00	5.17	3.33
MS-SLSQP	4.522	99.4	4.294	100.0	4.00	6.67	7.00

three values, we give $\omega = 0.75$ as a recommended starting point for applications, not as an optimized setting.

VIII. TESTS BEYOND THE BENCHMARK

This section tests whether the conclusions of Section VI hold for larger budgets, feasible starts, a real site and an external benchmark, i.e., where the limits of PSO-VNS lie.

A. Budget Scaling and Feasibility-Preserving Initialization

Budget. With 5 and 20 times the budget, PSO-VNS keeps the best average rank at all three budgets on the six largest benchmark cases (1.50, 1.17 and 2.17; Table VI; Fig. S9 of the supplement); the mean wake loss of PSO minus that of PSO-VNS is +0.10, +0.18 and +0.27 percentage points (per-case values in Table S14). The order of the other methods, however,

TABLE VII

HORNS REV 1 16-TURBINE BLOCK (WAKE-FREE AEP 149.26 GWh/YR): MEAN AEP (GWh/YR) OVER THE FEASIBLE RUNS OF 30 SEEDS AT 6,030 CALLS, FEASIBLE RUNS, HOLM-ADJUSTED WILCOXON p OF PSO-VNS VS. EACH METHOD, AND MEAN AEP LOSS (%) AT 6,030 CALLS WITH RANDOM (R) AND FEASIBILITY-PRESERVING (F) INITIALIZATION AND AT 30,030 AND 120,030 CALLS (R; 10 SEEDS AT 120,030); SUPERScript: FEASIBLE RUNS WHEN NOT ALL RUNS ARE FEASIBLE

Layout / method	6,030 calls, R			Loss (%)			
	AEP	Feas.	p_{Holm}	6k R	6k F	30k	120k
Installed layout	139.51	–	–	6.53	–	–	–
PSO-VNS	138.73	30/30	–	7.05	6.96	6.46	6.45
SSA-VNS	137.98	30/30	6.7×10^{-4}	7.56	6.99	7.16	6.48
LX-SSA-VNS	137.77	30/30	6.3×10^{-4}	7.70	7.06	7.06	6.51
RS-VNS	137.96	28/30	2.2×10^{-4}	7.57 ²⁸	n/r	6.98	6.24
VNS	137.67	30/30	2.2×10^{-4}	7.77	6.88	6.72	6.12
PSO	138.06	19/30	1.4×10^{-5}	7.51 ¹⁹	8.39	6.82 ²⁸	6.66
SSA	136.66	30/30	2.8×10^{-7}	8.44	7.89	8.14	7.61
LX-SSA	136.46	28/30	1.5×10^{-7}	8.58 ²⁸	7.96	8.48	8.36
DE	–	0/30	1.7×10^{-8}	–	8.39	7.28 ⁸	6.34
MS-SLSQP	136.80	30/30	1.5×10^{-7}	8.35	8.06	7.79	7.26

depends on the budget: PSO falls back from an average rank of 1.83 to 5.83, VNS (6.50 to 3.17) and DE move up, and at 120,030 evaluations VNS and SSA-VNS come close to PSO-VNS (3.17% and 3.23% vs. 3.14%). Only the first place of PSO-VNS is stable across budgets.

Feasible starts. With feasibility-preserving initialization, the share of feasible runs rises for SSA-VNS (95.0% to 100.0%), SSA (80.0% to 100.0%), LX-SSA (77.8% to 100.0%) and DE (1.1% to 100.0%). VNS alone is then best (VNS has the lowest mean wake loss, VNS the best average rank), and PSO-VNS ranks 2.83 on average (mean wake loss 3.94%, 4.08% with random starts; Table S12); RS-VNS was not run with this initialization. PSO and DE never improve on the best initial layout in these runs: since the turbine labels are arbitrary, velocity and difference moves that combine two feasible layouts almost always make turbines overlap, the feasibility-first comparison rejects them, and the global-best particle of PSO does not move. The single-coordinate moves of VNS are not affected, so a feasible start favors the VNS-based methods, whereas the penalty-guided search from random starts favors PSO: the advantage of PSO-VNS depends on the initialization.

B. Horns Rev 1 16-Turbine Block

Under our model (Table VII, Fig. 2), the installed layout, a regular 7D grid, reaches 139.51 GWh/yr (wake loss 6.53%). At 6,030 evaluations, PSO-VNS finds a feasible layout in all 30 runs and has the highest mean AEP, 138.73 GWh/yr (SD 0.83), significantly higher than that of every other method ($p_{Holm} \leq 6.7 \times 10^{-4}$). This mean is below the installed layout, which only the best run (140.39 GWh/yr) and 6 of the 30 runs exceed; of the other methods, only PSO (one run) does so. At 30,030 evaluations, the mean AEP of PSO-VNS (139.62 GWh/yr, loss 6.46%) exceeds the installed layout, as do 19 of the 30 runs; at 120,030 evaluations the loss is 6.45%, but VNS, RS-VNS and DE are lower, which confirms the budget dependence. PSO finds a feasible layout inside the parallelogram in only 19/30 runs from random starts (30/30

TABLE VIII

IEA37 CASE STUDY 1, 16- AND 36-TURBINE SCENARIOS: AEP IN MWH OF THE BEST RUN OF EACH METHOD AT 6,030 AND 30,030 CALLS (30 SEEDS), COMPARED WITH THE EXAMPLE LAYOUT AND THE BEST FEASIBLE PUBLISHED LAYOUT [20], [41]

Method / source	16 turbines ($r = 1300$ m)		36 turbines ($r = 2000$ m)	
	6,030	30,030	6,030	30,030
Example layout	366,941.6		737,883.1	
Best feasible published ^a	418,924.4		863,676.3	
PSO-VNS	402,782.4	413,331.9	787,708.9	827,385.0
PSO	403,753.1	412,702.4	787,988.7	830,256.4
SSA-VNS	403,251.3	406,924.6	766,803.3	819,930.5
SSA	395,150.2	401,459.3	750,707.4	802,538.2
LX-SSA	393,284.6	398,770.0	746,301.3	789,311.3
DE	357,391.3	397,402.7	–	–
VNS	400,169.3	404,076.0	785,791.3	824,230.2
MS-SLSQP	405,433.8	410,349.5	748,571.1	833,365.0

^aParticipant 4 in both scenarios (SNOPT with wake expansion continuation [TBD: verify against Baker *et al.* (2019)]). A higher submitted AEP places turbines up to 3.5 m outside the boundary and is not counted.

with feasible starts). The comparison uses our wake model and boundary; it does not assess the design of the real farm.

C. IEA37 Case Study 1

In IEA37 Case Study 1 the participants chose their own optimizers and budgets [20]. None of our methods reaches the best feasible published layouts (Table VIII, Fig. 2), which come from a gradient-based method with a budget of its own choice (footnote of Table VIII). At 30,030 evaluations, the relative AEP difference to them is -1.33% (16 turbines) and -4.20% (36 turbines) for the best PSO-VNS layout and -3.90% and -6.51% for the mean over the 30 runs; the best layout would rank third of 9 and seventh of 8 feasible submissions. The gap is larger for 36 turbines, where our best layout at 30,030 evaluations comes from the gradient-based MS-SLSQP. The metaheuristics compared here thus do not match the best gradient-based results; as the participants' budgets are not reported comparably, this comparison is indicative only.

D. Published Kusiak–Song Results and Computational Cost

Published Kusiak–Song results [5]–[8] are compared in Section S-VII of the supplement [TBD: pending author verification of the PDFs]. At equal numbers of evaluations, all methods need a similar time (0.24–1.82 ms per evaluation on the machines that produced the respective runs; Table S18); one 6,030-evaluation run of PSO-VNS takes 1.5–3.2 s, depending on N .

IX. SENSITIVITY TO THE MODELING ASSUMPTIONS

Power curve and wake model: We re-evaluated every feasible final layout of the main comparison, without re-optimization, with 1) a cubic power curve [42], $f_c(s) = P_{rated}(s^3 - s_{cut-in}^3)/(s_{rated}^3 - s_{cut-in}^3)$ between the cut-in and rated speeds, 2) the same curve with a 25 m/s cut-out, and 3) the Gaussian wake model of [43] with growth rate $K_G = 0.04$ and the superposition and Weibull scaling of the benchmark (Section S-V.A of the supplementary material). The absolute

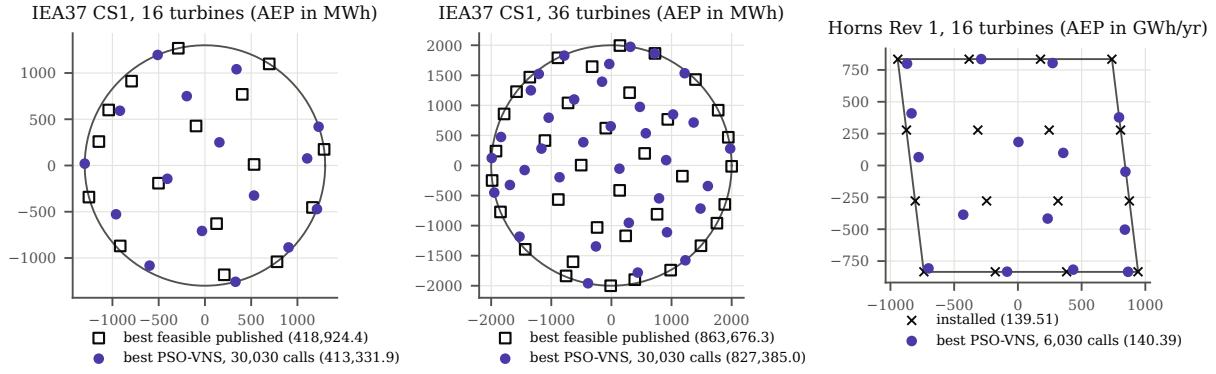


Fig. 2. Best layouts. Left and middle: IEA37 Case Study 1, 16 and 36 turbines, best PSO-VNS layout at the largest budget and best feasible published layout. Right: Horns Rev 1 16-turbine block, installed layout and best PSO-VNS layout at 6,030 calls.

TABLE IX
ROBUSTNESS TO THE BENCHMARK MODEL: ALL FEASIBLE FINAL LAYOUTS RE-EVALUATED (NOT RE-OPTIMIZED). Δ : MEAN RELATIVE CHANGE OF THE OBJECTIVE (%); RANK: AVERAGE RANK OF PSO-VNS OVER THE 68 CASES; BEST: BEST-RANKED METHOD; $\bar{\tau}$: MEAN KENDALL CORRELATION BETWEEN THE BENCHMARK AND ALTERNATIVE ORDERINGS WITHIN A CASE; SAME: CASES (%) WITH UNCHANGED BEST METHOD. THE RANKS OF ALL METHODS ARE GIVEN IN TABLE S19

Model	Δ (%)			Best	$\bar{\tau}$	Same
	DS I	DS II	Rank			
Benchmark	+0.0	+0.0	1.82	PSO-VNS	—	—
Cubic curve	-17.7	-36.9	1.82	PSO-VNS	0.95	94
Cubic, 25 m/s cut-out	-21.5	-37.0	1.80	PSO-VNS	0.95	93
Gaussian wake	-0.3	-0.4	1.97	PSO	0.66	54

energy depends strongly on the model (Table IX): the cubic curve gives 17.7% (Data Set I) and 36.9% (Data Set II) less expected power than the linear benchmark curve, so our energy values are benchmark values, not AEP predictions for a specific turbine. The ranking of the methods is stable under the cubic curves ($\bar{\tau} = 0.95$; same best method in 94% of the cases), and PSO-VNS keeps the best average rank. It is less stable under the Gaussian model ($\bar{\tau} = 0.66$; same best method in 54% of the cases), since layouts tuned to the sharp Jensen wake cone are judged differently by a smooth deficit: PSO ranks first (1.91) and PSO-VNS second (1.97), both ahead of all other methods. The small advantage of the VNS phase over PSO is thus specific to the wake model it optimizes, which is a limit of PSO-VNS.

Spacing and boundary handling: The optimized layouts often lie on the 4D constraint inherited from the benchmark: of the feasible final layouts of the eight methods, 49% also satisfy 5D and 36% satisfy 6D (4.1% and 0.1% for $N = 11-15$), so they cannot be transferred to sites with a larger spacing without re-optimization. At 5D and 6D, a multi-start packing search fits fewer turbines into each circle, and the largest 500-m cases may be infeasible (constructive lower bounds in Table S20; a re-optimization of six cases with SSA and LX-SSA is given in Table S21). The boundary rule also matters: an otherwise identical SSA that projects every turbine radially onto the circle instead of clipping the coordinates attains

higher mean benchmark objectives than the original SSA in all six representative cases at 3,030 evaluations (Mann-Whitney test, $p < 0.01$ in every case). We therefore apply the rule of Section II-C to all methods.

X. DISCUSSION AND RECOMMENDATIONS

The three findings share one implication: in continuous WFLOP, which method appears best depends as much on the comparison protocol as on the methods. The setting $w = 0.7$, $c_1 = c_2 = 2$ is a property of the PSO parameters, not of any particular study: it violates (9), and with it PSO ranks fifth of the eight methods of the main comparison (feasible in 86.0% of the runs), while SSA-VNS ranks first in the method pool of the previous study; with the constriction coefficients, PSO ranks second in the main comparison, PSO ranks first in that pool, and PSO is ahead of every salp-swarm method in both (Table I). An advantage over PSO is therefore informative only if the PSO baseline is convergent, and a swarm phase in a hybrid earns its place only if it beats a random-sampling control with the same budget, which the PSO phase does and the salp phases do not. PSO-VNS is a reasonable reference because it combines two standard components, adds one parameter (ω) and is feasible in every run; its gain over PSO, however, is not significant over all cases ($p = 0.30$) and appears mainly for $N \geq 10$.

For practitioners, this means that with random starts and a penalized objective, PSO-VNS (preferably with $\omega = 0.75$, Section VII-A) or a constriction PSO is a sound first choice, whereas VNS alone is competitive when a feasible start is available. Since the energy values are benchmark values, a layout should be re-evaluated, and preferably re-optimized, with the wake and power models of the site before it informs a decision. For researchers who propose WFLOP metaheuristics, we recommend the following, all of which are applied in this paper:

- 1) configure every baseline inside its convergence region (for PSO, check (9) or use (8)) and report all settings;
- 2) give all methods the same budget of objective calls, counting the initial population and every gradient or finite-difference call;

- 3) use seed-paired runs that start from the same initial population;
- 4) rank feasibility-aware: report the share of feasible runs and compare wake losses on feasible layouts only;
- 5) test both on the per-case means and on the runs of each case, with a multiplicity correction such as Holm's, and report non-significant results;
- 6) in ablations of hybrids, include a random-sampling control and the local method alone;
- 7) report feasible-start and model-robustness checks, and publish code and data.

XI. LIMITATIONS

The benchmark uses the top-hat Jensen wake, the linear Kusiak–Song power curve, circular farms, a $4D$ spacing and a prescribed N ; its energies are benchmark values, and when the final layouts are re-evaluated with a Gaussian wake, PSO ranks ahead of PSO-VNS. We did not re-optimize with the alternative models. The budget study reaches 120,030 evaluations but covers only the six largest cases and the Horns Rev 1 block, and the largest layouts have 36 turbines. The PSO and VNS settings are untuned defaults, and the pre-specified $\omega = 0.5$ is not the best split. With feasibility-preserving initialization, VNS ranks first, and PSO and DE do not move from the best initial layout, because velocity and difference moves between feasible layouts with arbitrary turbine labels make turbines overlap and are rejected by the feasibility-first comparison; our statements on PSO-VNS therefore refer to random starts. The only real site is a 16-turbine block of Horns Rev 1, and on IEA37 Case Study 1 the best PSO-VNS layouts still differ by -1.33% and -4.20% from the best feasible published layouts (reported for a gradient-based method, [TBD: verify against Baker et al. (2019)]). Our statement on the setting $w = 0.7$, $c_1 = c_2 = 2$ rests on the convergence theory and our benchmark; we did not re-run the comparisons of other studies. Finally, terrain, mixed turbine types, cables, grid connection, noise, environmental constraints and economic objectives such as the levelized cost of energy are outside our scope.

XII. CONCLUSION

This paper revisited metaheuristic comparisons for continuous WFLOP under a controlled protocol on 68 benchmark cases, a Horns Rev 1 block and IEA37 Case Study 1. First, the baseline configuration changes the conclusions: the PSO setting $w = 0.7$, $c_1 = c_2 = 2$ violates the convergence condition and places PSO behind salp-swarm hybrids, whereas the constriction setting places it ahead of every salp-swarm method. Second, in two-phase hybrids, the VNS phase improves every swarm, but the swarm phase helps only if the swarm is effective: the SSA and LX-SSA phases are statistically tied with random sampling, the Laplace step gives no gain, and a PSO phase beats random sampling, VNS alone and the salp phases. Third, PSO-VNS is a simple and consistent reference method: it is feasible in every run, has the best average rank (1.82) and the best rank at all three budgets, is better than PSO for $N \geq 10$, and its best Horns Rev 1 layout exceeds the installed one; yet it is not significantly better than

PSO over all cases, VNS alone is best from feasible starts, PSO ranks first under the Gaussian re-evaluation, and on IEA37 it stays below the best published layouts. Future work will study a stagnation-triggered switch between the phases and extend the evaluation to larger farms, higher-fidelity wake models and economic objectives.

ACKNOWLEDGMENT

[TBD: acknowledgment, or delete this section]

DATA AVAILABILITY

The following are openly available at [TBD: Zenodo DOI]:

- 1) the source code of all methods and of the benchmark evaluator;
- 2) the per-run results of all studies, i.e., for every run the method, case, seed, budget, initialization, number of evaluations, final objective, feasibility, final turbine coordinates and convergence curve, including the runs with the earlier PSO setting; and
- 3) the scripts that implement the evaluation protocol (seed-paired runs, feasibility-aware ranking, case-mean and run-level tests with Holm correction, and the re-evaluation with the alternative power curve and wake model). One command (`sh analysis/build_mpce_paper.sh`) regenerates every table, figure and number of this paper and of its supplementary material from the per-run results. The supplementary material contains the model details, the pseudocode of SSA and LX-SSA, the wind data, the per-case results of all studies, the coordinates of the best layouts and further figures.

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